Wireless Communications Homework II

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Exercise *(Doppler Shift Simulation and Theoretical Analysis)*

In a wireless mobile communication system, when a mobile station (MS) moves with velocity v and receives an unmodulated carrier frequency f_c, the observed Doppler shift is:

$$f_D(t) = f_m \cos\theta(t) \tag{1}$$

where $f_m = \frac{vf_c}{c}$ is the maximum Doppler frequency, and $\theta(t) \sim \text{Uniform}(-\pi, \pi)$ is the incoming wave's incidence angle.

Simulation Parameters & Key Code:
Monte Carlo simulations are performed using $N = 1,111,111$ sampling points. The core implementation steps are summarized below:
Parameters Setup & Doppler Evaluation:

```
N = 1111111; c = 3e8;
v = 20 * (1000/3600); fc = 2e9;  % Example for (a)
fm = v * fc / c;

theta = -pi + 2*pi*rand(N, 1);
f_D = fm * cos(theta);
```

Statistical Visualization:

```
histogram(f_D, 50, 'Normalization', 'pdf');
ecdf(f_D);
```

(a) Simulation for $v = 20$ km/hr and $f_c = 2$ GHz

System Parameters:

- Velocity $v = 20$ km/hr ≈ 5.556 m/s
- Carrier frequency $f_c = 2$ GHz
- Maximum Doppler frequency $f_m \approx 37.04$ Hz

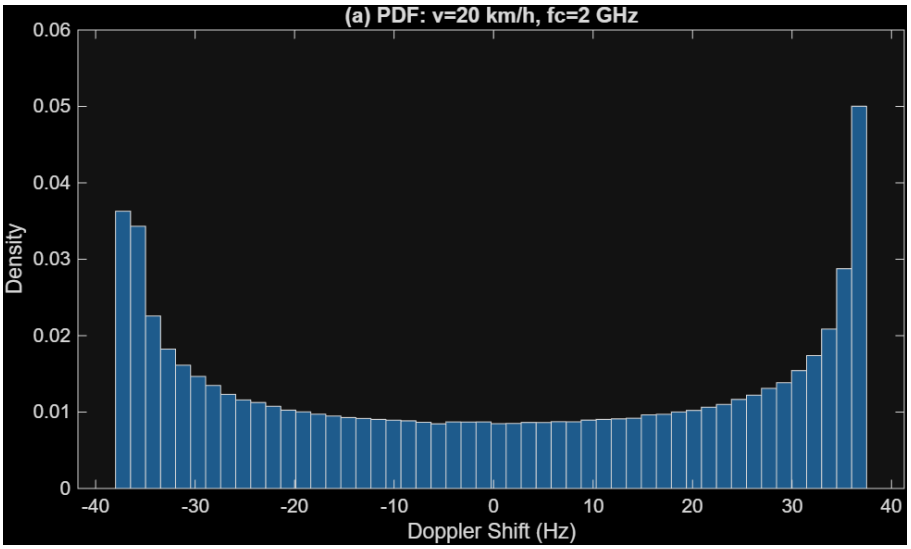


Figure 1: Simulated PDF of Doppler shift for $v = 20$ km/hr and $f_c = 2$ GHz.

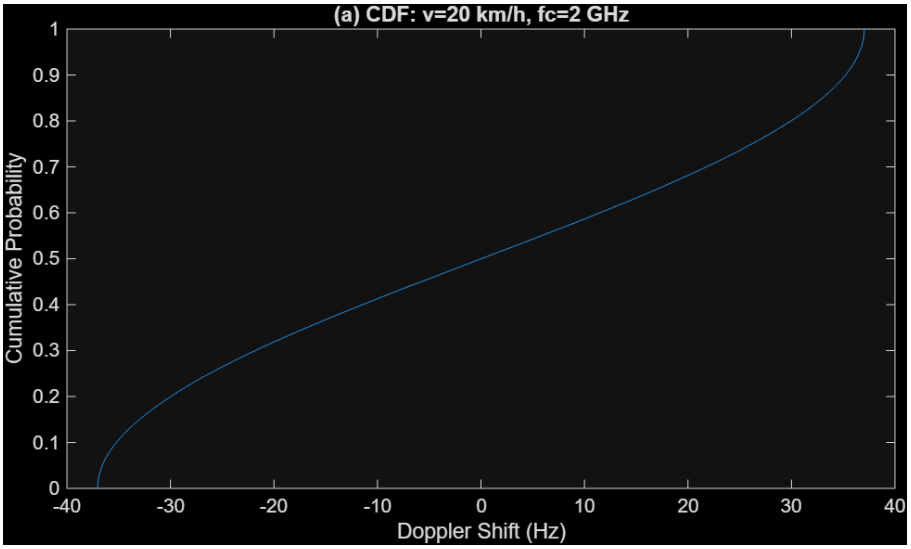


Figure 2: Simulated CDF of Doppler shift for $v = 20$ km/hr and $f_c = 2$ GHz.

(b) Simulation for $v = 90$ km/hr and $f_c = 26$ GHz

System Parameters:

- Velocity $v = 90$ km/hr $= 25$ m/s
- Carrier frequency $f_c = 26$ GHz
- Maximum Doppler frequency $f_m \approx 2166.67$ Hz

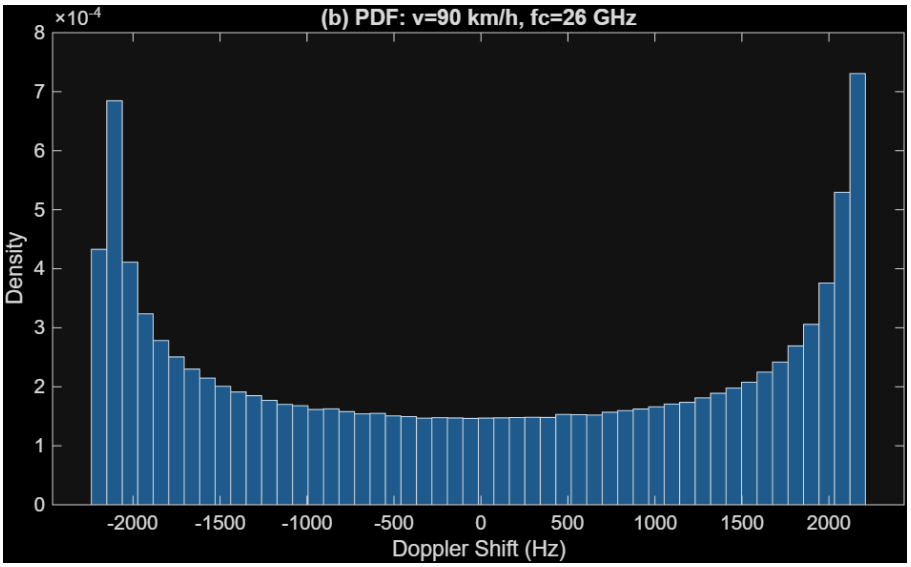


Figure 3: Simulated PDF of Doppler shift for $v = 90$ km/hr and $f_c = 26$ GHz.

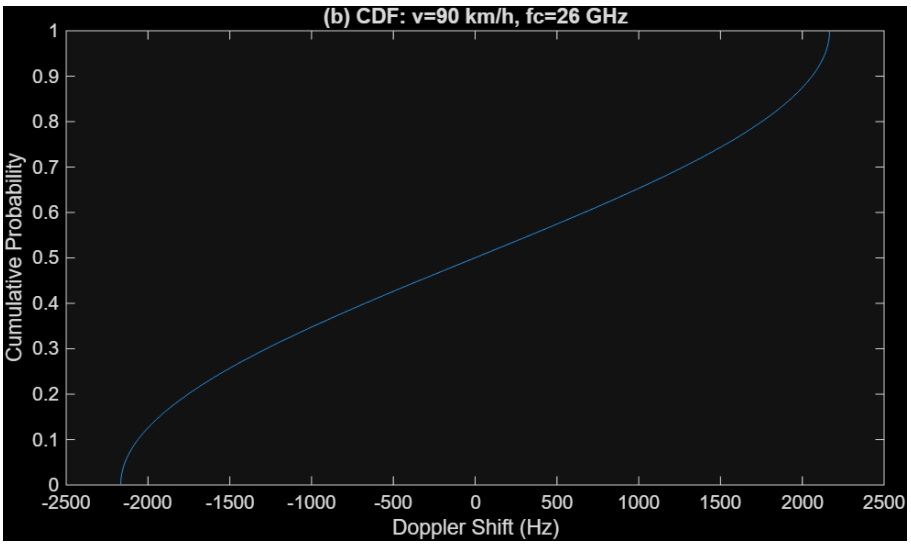


Figure 4: Simulated CDF of Doppler shift for $v = 90$ km/hr and $f_c = 26$ GHz.

(c) Simulation for $f_c = 2$ GHz and $v \sim$ Uniform$(20, 90)$ km/hr

System Parameters:

- Carrier frequency $f_c = 2$ GHz
- Velocity v is uniformly distributed between 20 km/hr and 90 km/hr

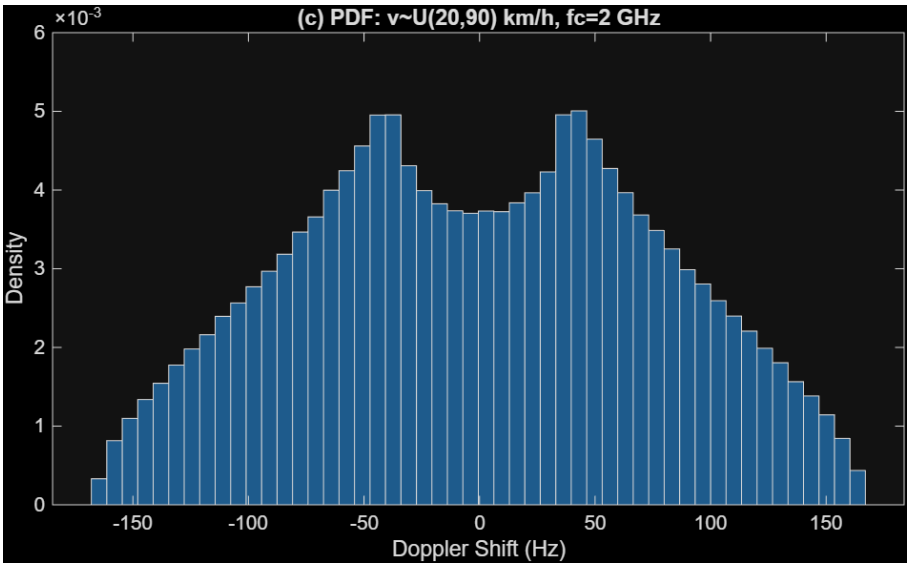


Figure 5: Simulated PDF of Doppler shift under random velocity $v \sim$ Uniform$(20, 90)$ km/hr.

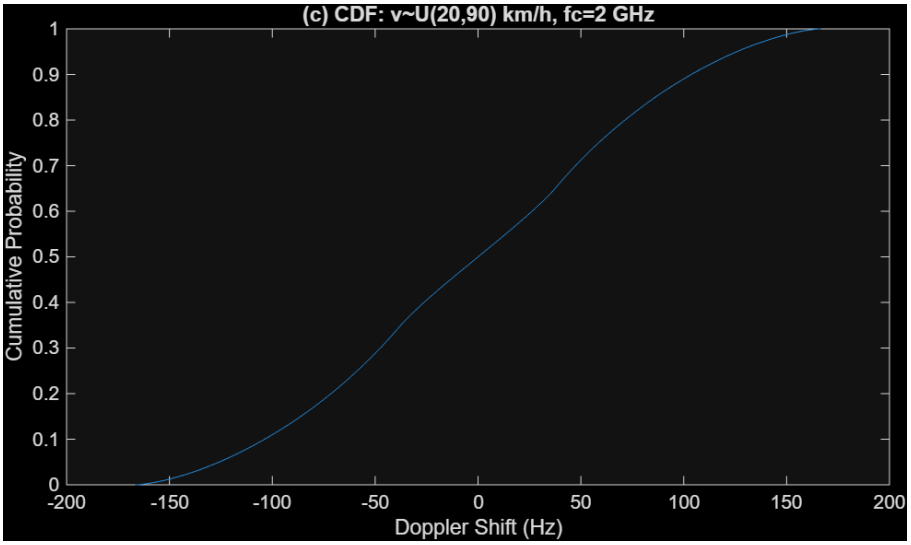


Figure 6: Simulated CDF of Doppler shift under random velocity $v \sim$ Uniform$(20, 90)$ km/hr.

(d) Theoretical Derivation and Comparison

Theoretical Derivation:
Given a fixed velocity v and carrier frequency f_c, let the observed Doppler shift be a random variable $Y = f_D = f_m \cos\Theta$, where Θ is the incoming wave's incidence angle.

According to the problem description, Θ is uniformly distributed between $-\pi$ and π, so its probability density function (PDF) is:

$$p_\Theta(\theta) = \frac{1}{2\pi}, \quad -\pi \le \theta \le \pi \tag{2}$$

1. Deriving the Probability Density Function (PDF):
To find the PDF of $Y = g(\Theta) = f_m \cos\Theta$, we use the fundamental theorem of transformation for continuous random variables. The transformation function has two roots for any given $y \in (-f_m, f_m)$ within the domain $[-\pi, \pi]$:

$$\theta_1 = \arccos\left(\frac{y}{f_m}\right) \quad \text{and} \quad \theta_2 = -\arccos\left(\frac{y}{f_m}\right) \tag{3}$$

The derivative of the transformation function is:

$$\frac{dy}{d\theta} = -f_m \sin\theta \tag{4}$$

Applying the transformation formula by summing the contributions from all roots:

$$p_Y(y) = \sum_i \frac{p_\Theta(\theta_i)}{\left|\frac{dy}{d\theta}\big|_{\theta=\theta_i}\right|} = \frac{p_\Theta(\theta_1)}{f_m|\sin\theta_1|} + \frac{p_\Theta(\theta_2)}{f_m|\sin\theta_2|} \tag{5}$$

Since $\sin\left(\arccos\left(\frac{y}{f_m}\right)\right) = \sqrt{1-\left(\frac{y}{f_m}\right)^2}$ and $p_\Theta(\theta) = \frac{1}{2\pi}$, we get:

$$p_Y(y) = 2 \cdot \frac{\frac{1}{2\pi}}{f_m\sqrt{1-\left(\frac{y}{f_m}\right)^2}} = \frac{1}{\pi f_m\sqrt{1-\left(\frac{y}{f_m}\right)^2}}, \quad -f_m < y < f_m \tag{6}$$

2. Deriving the Cumulative Distribution Function (CDF):
The CDF is obtained by integrating the theoretical PDF from the lower bound $-f_m$ to y:

$$F_Y(y) = P(Y \le y) = \int_{-f_m}^{y} p_Y(u)\,du = \int_{-f_m}^{y} \frac{1}{\pi f_m\sqrt{1-\left(\frac{u}{f_m}\right)^2}}\,du \tag{7}$$

By using the substitution $u = f_m \sin\phi$ (where $du = f_m\cos\phi\,d\phi$), the integral becomes:

$$F_Y(y) = \int_{-\pi/2}^{\arcsin(y/f_m)} \frac{1}{\pi f_m\cos\phi} \cdot f_m\cos\phi\,d\phi = \frac{1}{\pi}\int_{-\pi/2}^{\arcsin(y/f_m)} d\phi \tag{8}$$

Evaluating the definite integral yields the theoretical CDF:

$$F_Y(y) = \frac{1}{\pi}\left[\arcsin\left(\frac{y}{f_m}\right) - \left(-\frac{\pi}{2}\right)\right] = \frac{1}{2} + \frac{1}{\pi}\arcsin\left(\frac{y}{f_m}\right), \quad -f_m \le y \le f_m \tag{9}$$

Simulation and Comparison:
Using the parameters from part (a) ($v = 20$ km/hr, $f_c = 2$ GHz, and $f_m \approx 37.04$ Hz), $N = 1,111,111$ Monte Carlo samples are evaluated against the theoretical derivations.

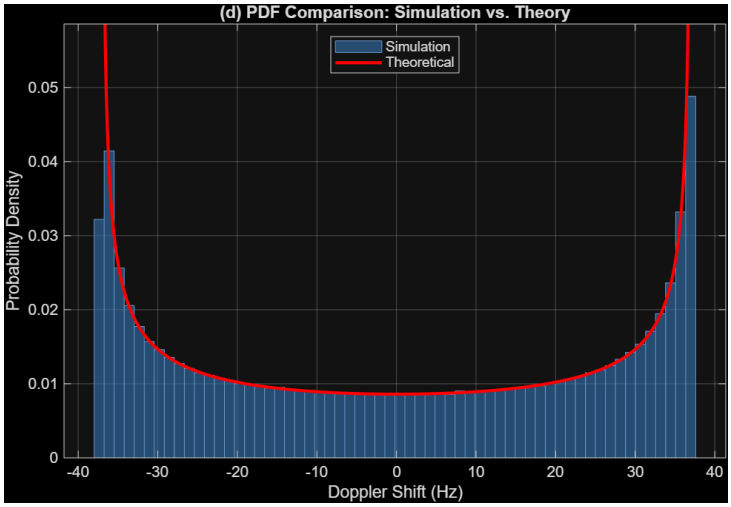


Figure 7: Comparison between Monte Carlo simulation results and the theoretical PDF for fixed $v = 20$ km/hr and $f_c = 2$ GHz.

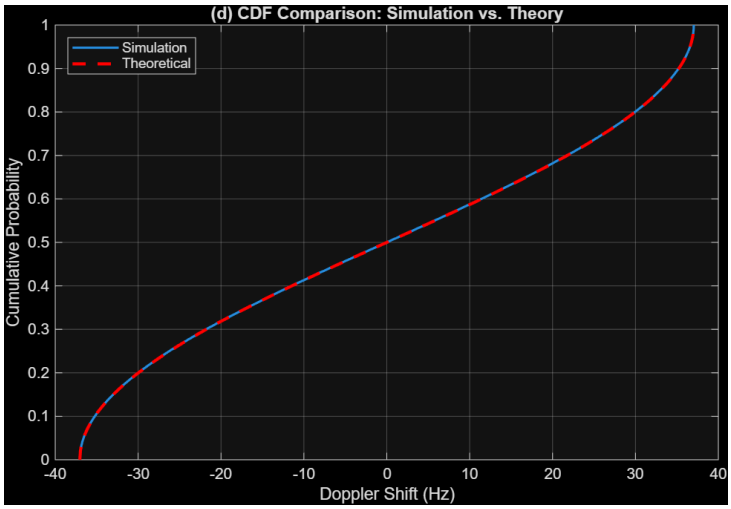


Figure 8: Comparison between Monte Carlo simulation results and the theoretical CDF for fixed $v = 20$ km/hr and $f_c = 2$ GHz.

Discussion and Conclusion:
As you can see in Fig. 7, the simulated PDF nicely captures the classic "bathtub" shape of Jakes' spectrum, peaking heavily near $\pm f_m$ while staying relatively flat in the middle, which matches the theoretical curve quite well. On top of that, Fig. 8 shows that the empirical ECDF overlaps almost perfectly with the theoretical CDF. Overall, these results strongly confirm that both our theoretical math and our Monte Carlo simulation are spot on.